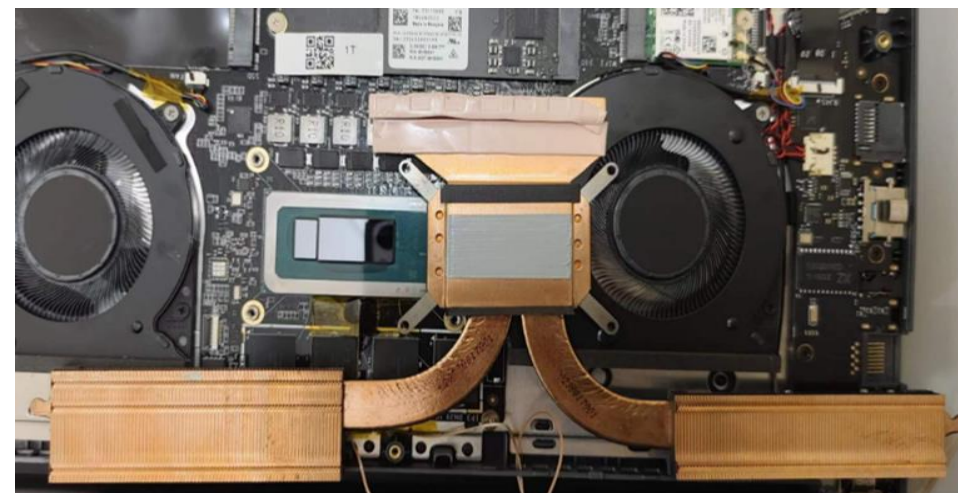
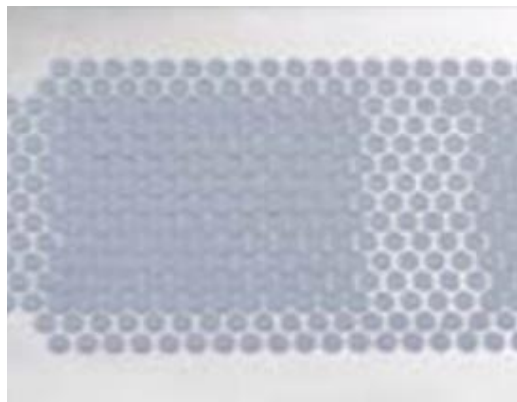
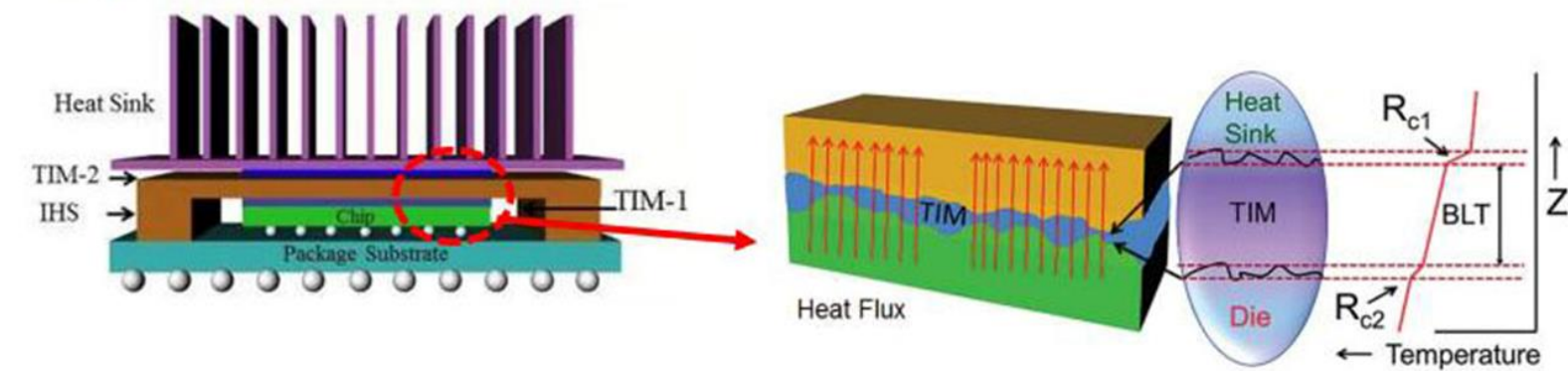


Deep-Materials Two Part Liquid Gap Filler Products



Use fundamental science to make practical improvements in TIM

Deep-Materials

TIM Product Brochure

Use *fundamental science* to make practical improvements in Thermal Materials

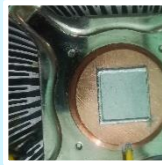
CPU/GPU TIMs



Grease

- Best in class performance
- Significantly reduce pump out

Highest performance traditional grease & Innovative Liquid Metal grease with low resistance



Liquid Metal

- Multi-phase Liquid metal – high viscosity
- Practical to use
- Highest performance

Brand new category of TIM with perf. of LM but ease of grease

Gap Fillers



Gels

- Highest conductivity
- Almost no oil seepage

Many kinds of unique gels – e.g. ultra low oil seepage gel, low dielectric constant gels



Gap Pads

- Almost eliminate oil seepage
- Reliable to very high temperatures

Ultra low oil seepage & ultra low outgassing silicone pads

GAP Fillers

- For Applications in lower power/thicker TIM applications
- Do not require high pressure or mechanical support
- Typically used in consumer electronics, SSDs, Radios
- Performance measured in thermal conductivity (W/mK)

Our offerings in this space

- Liquid Gap Fillers/Gels
 - Single part (Fully cured)/Two part (in situ cure)
- Gap pads

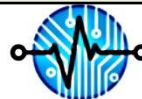
Two Parts Liquid Gap Filler



Applications:
Consumer Electronics, Automotive, Computing
SSDs, Memory Modules, Networking, Data Centers

	TPGF-180-LD	TPGF-200	TPGF-200LP-RW	TPGF-300-RW	TPGF-350	TPGF-500	Testing Method
Color A/B	Red/White	Yellow/White	Pink/White	Yellow/White	Blue/White	Green/White	Visual
Mixing Ratio	1: 1	1: 1	1: 1	1: 1	1: 1	1: 1	—
Viscosity (cps)	150,000	170,000	250,000	270,000	250,000	280,000	ASTM D2196
Density (g/cc)	1.8	2.5	1.8	3.0	3.2	3.3	ASTM D792
Hardness (shore 00, after fully cured)	50	45	50	45	50	60	ASTM D2240
Volatility (%)	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	ASTM E595
Operation Temperature (°C)	-50~150	-50~200	-50~200	-50~200	-50~200	-50~200	—
Electrical Parameters							
Breakdown Voltage (Kv/mm)	>6.0	>8.0	>8.0	>8.0	>8.0	>8.0	ASTM D149
Dielectric Constant (1GHz)	7.0	6.8	3.7	7.5	7.5	8.0	ASTM D150
Volume Resistance (Ω.cm)	10 ¹³	10 ¹³	10 ¹³	10 ¹³	10 ¹³	10 ¹³	ASTM D257
Flammability rating	V-0	V-0	V-0	V-0	V-0	V-0	UL 94
Thermal Performance							
Thermal Conductivity (W/m-K)	1.8	2.0	2.0	3.0	3.5	5.0	ASTM D5470
Curing Parameters							
Operation Time@ 25°C (min)	120	120	120	120	120	120	-
Curing time@ 25°C (Hrs)	24	24	24	24	24	24	-
Curing Time@ 100°C (min)	10	10	10	10	10	10	-
SPECIAL PROPERTIES	-	-	Easy peel without residue (reworkable)		-	-	-

-RW line of products can be used as a dispensed 3D gap pads.
Contact us for more details.



Deep Materials

Contact Us

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